

CHORDS AND ARCS

Unit

27

• Weightage = 5%

Students Learning Outcomes (SLOs)

After completing this unit, students will be able to:

- Understand the following theorems along with their corollaries and apply them to solve allied problems.
 - ❖ If two arcs of a circle (or of congruent circles) are congruent, then the corresponding chords are equal.
 - ❖ If two chords of a circle (or of congruent circles) are equal, then their corresponding arcs (minor, major, semi-circular) are congruent.
 - ❖ Equal chords of a circle (or of congruent circles) subtend equal angles at the centre (at the corresponding centres).
 - ❖ If the angles subtended by the two chords of a circle (or congruent circles) at the centre (corresponding centres) are equal, the chords are equal.



Introduction:

In this unit, we discuss the theorems related to chords and arcs of a circle.

Theorem 27.1

If two arcs of a circle (or congruent circles) are congruent then the corresponding chords are equal.

We shall prove the theorem:

- For one circle
- For two congruent circles

i. For one circle

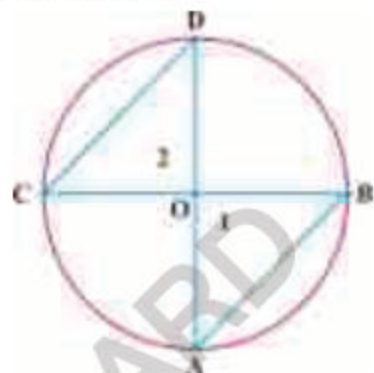
Given: A circle with center O whose $\widehat{AB}$ and $\widehat{CD}$ are congruent arcs i.e. $\widehat{AB} \cong \widehat{CD}$.

$\overline{AB}$ and $\overline{CD}$ are the corresponding chords of the given congruent arcs.

To prove: $\overline{AB} \cong \overline{CD}$

Construction: Join the points O with A, B, C and D.

Proof:



Statement	Reason
In $\triangle OAB \leftrightarrow \triangle OCD$	
$\overline{OA} \cong \overline{OC}$	Radii of same circle
$\overline{OB} \cong \overline{OD}$	Radii of same circle
$m\angle 1 = m\angle 2$	Central angles of two congruent arcs
$\triangle OAB \cong \triangle OCD$	S.A.S postulate
$\overline{AB} \cong \overline{CD}$	Corresponding sides of congruent triangles

Q.E.D

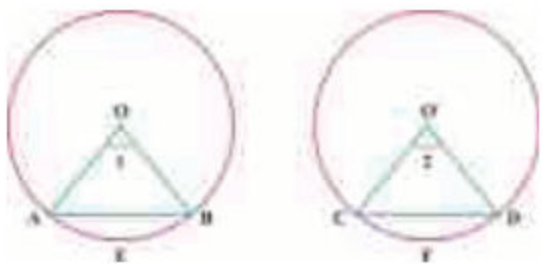
ii. For two congruent circles

Given: Two congruent circles with centers O and O' respectively. $\widehat{AB}$ and $\widehat{CD}$ are congruent arcs of these circles where $\overline{AB}$ and $\overline{CD}$ are the corresponding chords.

To prove: $\overline{AB} \cong \overline{CD}$

Construction: Join O to A and B. Join O' to C and D.

Proof:



Statement	Reason
In $\triangle OAB \leftrightarrow \triangle O'CD$	
$\overline{OA} \cong \overline{O'C}$	Radii of same circle
$\overline{OB} \cong \overline{O'D}$	Radii of same circle
$m\angle 1 = m\angle 2$	Central angles of two congruent arcs
$\therefore \triangle OAB \cong \triangle O'CD$	S.A.S postulate
$\therefore \overline{AB} \cong \overline{CD}$	Corresponding sides of two congruent triangle



Theorem 27.2 (Converse of theorem 1)

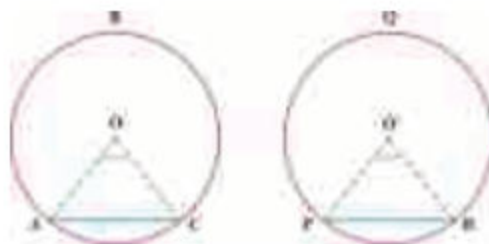
If two chords of a circle (or of congruent circles) are equal, then their corresponding arcs (minor, major, semi-circle) are congruent.

Given: Two congruent circles with center O and O' respectively having two congruent arcs i.e., $\widehat{AC} \cong \widehat{PR}$.

To prove: $\widehat{AC} \cong \widehat{PR}$

Construction: Join O with A and C , O' with P and R .

Proof:



Statement	Reason
In $\triangle AOC \leftrightarrow \triangle PO'R$	
$\overline{OA} \cong \overline{O'P}$	Radii of congruent circles
$\overline{OC} \cong \overline{O'R}$	Radii of congruent circles
$\widehat{AC} \cong \widehat{PR}$	Given
$\triangle OAC \cong \triangle PO'R$	S.S.S $\cong$ S.S.S
$m\angle AOC = m\angle PO'R$ (i)	Corresponding angles of congruent $\triangle$ s.
Hence $\widehat{AC} \cong \widehat{PR}$	From eq: (i)

Q.E.D

Example 1:

A point P on the circumference of a circle is equidistant from the radial segments $\overline{OA}$ and $\overline{OB}$. Prove that $m\widehat{AP} = m\widehat{BP}$

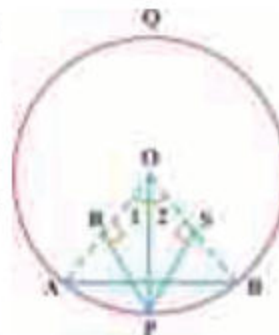
Solution:

Given: $\overline{AB}$ is the chord of a circle with center O . Point P on the circumference of the circle is equidistant from the radial segment $\overline{OA}$ and $\overline{OB}$. i.e. $m\overline{PR} = m\overline{PS}$

To prove: $m\widehat{AP} = m\widehat{BP}$

Construction: Join O with P . Write $m\angle 1$ and $m\angle 2$ as shown in the figure.

Proof:



Statement	Reason
In right angle $\triangle OPR \leftrightarrow \triangle OPS$	
$m\overline{OP} = m\overline{OP}$	Common
$m\overline{PR} = m\overline{PS}$	
$\therefore \triangle OPR \cong \triangle OPS$	H.S $\cong$ H.S
So $m\angle 1 = m\angle 2$ (i)	Corresponding angles of congruent $\triangle$ s
$\Rightarrow \widehat{AP} \cong \widehat{BP}$	From (i)
Hence $m\widehat{AP} = m\widehat{BP}$	By definition of congruent arcs



Theorem 27.3

Equal chords of a circle (or of congruent circles) subtend equal angles at the center (at the corresponding centers).

We shall prove theorem:

i) For one circle

ii) For two circles

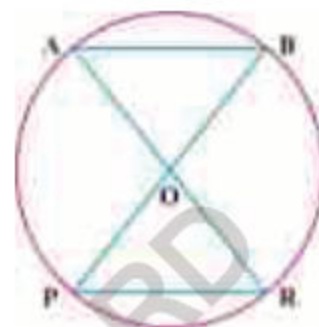
i) For one circle

Given: Circle with centre O, having two congruent chords i.e., $\overline{AB} \cong \overline{PR}$.

To prove: $m\angle AOB = m\angle POR$

Construction: Join O with A and B and also O with P and R.

Proof:



Statement	Reason
$\overline{AB} \cong \overline{PR}$ (i)	Given
$\overline{OB} \cong \overline{OR}$ (ii)	$\overline{OB}, \overline{OR}, \overline{OA}$ and $\overline{OP}$ are radii of same circle
and $\overline{OA} \cong \overline{OP}$ (iii)	
$\triangle AOB \cong \triangle POR$	S.S.S $\cong$ S.S.S
$m\angle AOB = m\angle POR$	Corresponding angles of congruent triangles

Q.E.D

ii) For two congruent circles

Given:

Two congruent circles with center O and O' respectively so that $m\widehat{AC} = m\widehat{PR}$. Where $\widehat{AC}$ and $\widehat{PR}$ are the chords of circles.

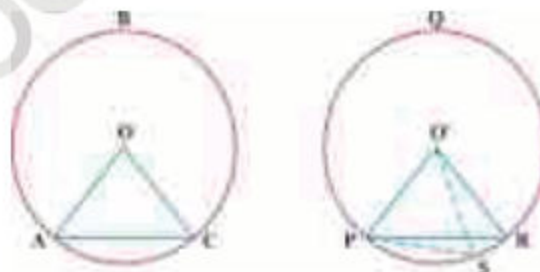
To prove:

$$m\angle AOC = m\angle PO'R$$

Construction:

Take a point S on arc PR and join S to P and O' such that $\angle AOC \cong \angle PO'S$.

Proof:



Statement	Reason
Let $m\angle AOC = m\angle PO'R$	
But $m\angle AOC \cong m\angle PO'S$	Construction
$\therefore \widehat{AC} \cong \widehat{PS}$ (i)	Arcs subtended by equal central angles of congruent circles
$\therefore m\widehat{AC} = m\widehat{PS}$ (ii)	From theorem (1)
But $m\widehat{AC} = m\widehat{PR}$ (iii)	Given
$\therefore m\widehat{PR} = m\widehat{PS}$	Using (ii) and (iii)
Which is only possible, if R coincides with S.	Because P is common point
$\Rightarrow m\angle AOC = m\angle POR$	Our supposition is working



Corollary 1:

In congruent circles or in same circle, if central angles are equal then corresponding sectors are equal.

Corollary 2:

In congruent circles or in same circle, arcs will subtend unequal central angles for unequal chords.

Example 1:

Prove that the internal bisector of a central angle in a circle bisects the corresponding arc.

Given: A circle with center O. $\overline{OP}$ is an internal bisector of central angle AOB. i.e. $\angle 1 \cong \angle 2$

To prove: $\widehat{AP} \cong \widehat{BP}$

Construction: Draw $\overline{AP}$ and $\overline{BP}$, then write $\angle 1$ and $\angle 2$ are corresponding angles of $\widehat{AP}$ and $\widehat{BP}$ respectively.

Proof:



Statement	Reason
In $\triangle OAP \leftrightarrow \triangle OBP$	
$\overline{OA} = \overline{OB}$	Radii of same circle
$m\angle 1 = m\angle 2$	Given
and $\overline{OP} = \overline{OP}$	Common
$\therefore \triangle OAP \cong \triangle OBP$	S.A.S postulate
Hence $\overline{AP} \cong \overline{BP}$	Corresponding sides of congruent $\triangle$ s
$\Rightarrow \widehat{AP} \cong \widehat{BP}$	Arcs corresponding to equal chords in a circle.

Q.E.D

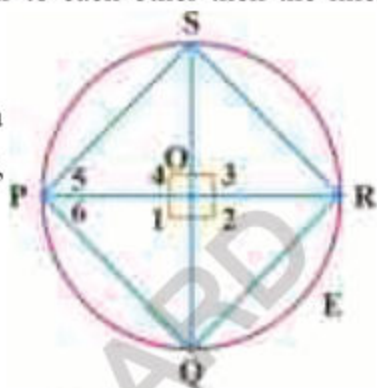


Example 2:

Prove that in a circle, if two diameters are perpendicular to each other then the lines joining their ends form a square.

Given:

Given $\overline{PR}$ and $\overline{QS}$ are two perpendicular diameters of a circle with center O . So $PQRS$ is a quadrilateral. Naming $\angle 1, \angle 2, \angle 3, \angle 4, \angle 5$ and $\angle 6$ as shown in figure.



To prove:

$PQRS$ is a square.

Proof:

Statement	Reason
$\overline{PR}$ and $\overline{QS}$ are two perpendicular diameters of a circle with center O .	Given
$m\angle 1 = m\angle 2 = m\angle 3 = m\angle 4 = 90^\circ$	Diameters are perpendicular to each other
$m\overline{PQ} = m\overline{QR} = m\overline{RS} = m\overline{SP}$ (i)	Arcs opposite to the equal central angles in a circle
$m\overline{PQ} = m\overline{QR} = m\overline{RS} = m\overline{SP}$ (ii)	Chords corresponding to equal arcs.
In right $\triangle POS$,	
$m\overline{PO} = m\overline{OS}$ (iii)	$\overline{PO}$ and $\overline{OS}$ are radii of circle
$\triangle POS$ is right isosceles, triangle	From (iii)
$m\angle 5 = 45^\circ$ (iv)	$\triangle POS$ is right isosceles, triangle
Similarly, $\triangle POQ$ is right isosceles triangle	By the above process
$m\angle 6 = 45^\circ$ (v)	
Moreover	
$m\angle P = m\angle 5 + m\angle 6$	Angle addition postulate
$m\angle P = 45^\circ + 45^\circ$	By using (iv) and (v)
$m\angle P = 90^\circ$ (vi)	
Similarly, $m\angle Q = m\angle R = m\angle S = 90^\circ$ (vii)	By the above process
$PQRS$ is a square	Using eq (ii), (vi) and (vii)

Q.E.D



Theorem 27.4

If the angles subtended by two chords of a circle (or congruent circles) at the centres (corresponding centres) are equal, the chords are equal.

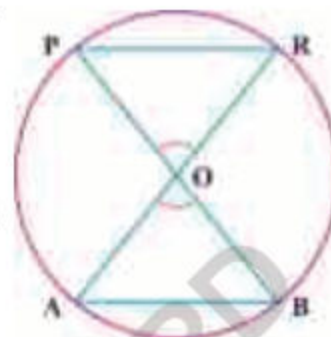
(a) For one circle

Given:

Circle with centre O, having chords $\overline{PR}$ and $\overline{AB}$. The angles subtended by both arcs are congruent i.e. $\angle POR \cong \angle AOB$.

To prove: $m\overline{AB} \cong m\overline{PR}$

Construction: Join O with P and R, and also with A and B.



Proof:

Statement	Reason
$\overline{OR} \cong \overline{OB}$ (i)	Radius of same circle
and $\overline{OP} \cong \overline{OA}$ (ii)	Radii of same circle
$\angle POR \cong \angle AOB$ (iii)	Given
$\triangle POR \cong \triangle AOB$	S.A.S $\cong$ S.A.S (Postulate)
$\overline{AB} \cong \overline{PR}$	Corresponding side of congruent triangles

Q.E.D

(b) For two congruent circles

Given:

Two congruent circles with center O and O' with $\overline{AC}$ and $\overline{PR}$ are chords. Subtending equal angles at centre i.e. $\angle AOC \cong \angle PO'R$.

To prove: $m\overline{AC} = m\overline{PR}$

Proof:

Statement	Reason
In $\triangle OAC \leftrightarrow \triangle O'PR$	
$\overline{OA} \cong \overline{O'P}$	Radii of congruent circles
$\angle AOC \cong \angle PO'R$	Given
$\overline{OC} \cong \overline{O'R}$	Radii of congruent circle
$\therefore \triangle OAC \cong \triangle O'PR$	S.A.S postulate
Hence $m\overline{AC} = m\overline{PR}$	Corresponding sides of congruent $\triangle$ s

Q.E.D

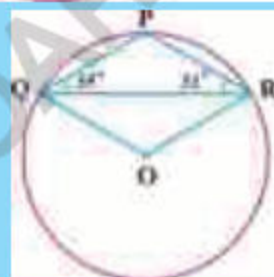


EXERCISE 27.1

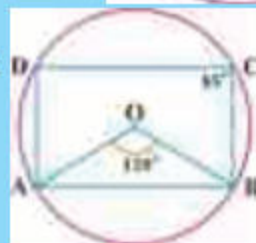
1. In a circle prove that the arcs between two parallel and equal chords or equal.
2. Prove that in equal circles, equal central angles have equal arcs.
3. In the following figure, O is the centre of the circle. Find the value of x and y .



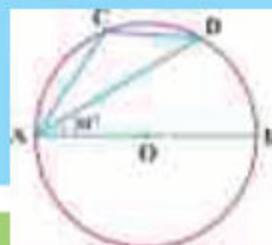
4. In the following figure $m\angle PQR = 35^\circ$ and $m\angle PRQ = 31^\circ$. Find $m\angle QPR$ and $m\angle OQR$.



5. In the figure O is the center of circle, given that $m\angle AOB = 120^\circ$ and $m\angle BCD = 85^\circ$. Find $m\angle OAD$



6. In the following figure, $\overline{AB}$ is diameter of the circle given that $m\angle DAB = 30^\circ$. Find $m\angle ACD$



REVIEW EXERCISE 27

1. Multiple Choice Question:

Tick the correct option.

- i. If a chord of a circle subtends a central angle of 60° . Then the chord and the radial segment are _____.
 (a) Parallel (b) Perpendicular (c) Congruent (d) Incongruent
- ii. An arc subtends a central angle of 45° then the corresponding chords will subtend a central angle of ?
 (a) 15° (b) 30° (c) 45° (d) 60°
- iii. A pair of chords of a circle subtending two congruent central angle are.
 (a) Perpendicular (b) Non congruent (c) Congruent (d) None of these



- iv. The arcs opposite to congruent central angles of a circle are always.
(a) Parallel (b) Congruent (c) Perpendicular (d) None of these
- v. A 6cm long chord subtends a central angle of 60° . The radial segment of this circle is
(a) 4cm (b) 6cm (c) 5cm (d) 8cm
- vi. The chord length of a circle subtending a central angle of 180° is always
(a) equal to the radial segment
(b) less than radial segment
(c) double of radial segment
(d) half of the radial segment.
- viii. The length of a chord and the radial segment of a circle are congruent, the central angle made by the chord will be:
(a) 60° (b) 75° (c) 90° (d) 45°
- ix. Out of two congruent arcs of a circle, if one arc makes a central angle of 30° then the other arc will subtend the central angle:
(a) 60° (b) 90° (c) 75° (d) 30°
- x. Diameter divides the circle into _____ parts.
(a) Two (b) Three (c) Four (d) All of these

SUMMARY

- The circles are congruent if their radii are equal.
- Equal chords of a circle subtend equal angles at the centre.
- The straight line joining any two points of the circumference is called a chord of the circle.
- The portion of a circle bounded by an arc and a chord is known as the segment of a circle.
- The boundary traced by a moving point in a circle is called its circumference.
- A portion of the circumference of a circle is called an arc of the circle.
- A straight line, drawn from the centre of a circle bisecting a chord is perpendicular to the chord.
- A perpendicular drawn from the centre of a circle on a chord, bisects the chord.
- If two arcs of a circle (or of congruent circles) are congruent, then the corresponding chords are equal.
- If two chords of a circle (or of congruent circles) are equal, then their corresponding arcs (minor, major, semi circular) are congruent.
- Equal chords of a circle (or of congruent circles) subtend equal angles at the centre (at the corresponding centres).
- If the angle subtended by two chords of a circle (or congruent circles) at the centre (corresponding centres) are equal, the chords are equal.